

RRC_Raptor_V4H

QSPI Flash

- Introduction

V0.0.1

Contents

LIST OF FIGURES	3
LIST OF TABLES	4
1 VERSION LIST.....	5
2 HASHTAG	6
1. Overview	7
1.1. Serial Flash boot at single read 40MHz using DMA.....	7
1.2. Serial Flash boot at DDR read 80MHz using DMA	8
2. Setup Step	10
2.1. Serial Flash boot at single read 40MHz using DMA.....	10
2.1.1. Setting boot parameter in sa0.c.....	10
2.1.2. Following qspi_sdr_transfer_mode() in rpc.c.....	10
2.1.3. Pull-up/down settings.....	11
2.2. Serial Flash boot at DDR read 80MHz using DMA	11
2.2.1. Setting boot parameter in sa0.c.....	11
2.2.2. Following rpc_init () in rpc.c.....	12
2.2.3. Following qspi_ddr_transfer_mode_quad_io_read () in rpc.c....	13
2.2.4. Pull-up/down settings.....	14
3. References.....	15

LIST OF FIGURES

FIGURE 1-1. EXTERNAL ADDRESS SPACE READ MODE INITIAL FLOW.....	8
FIGURE 1-2. MANUAL OPERATING MODE INITIAL FLOW.	8
FIGURE 1-3. MANUAL OPERATING MODE DATA TRANSFER SETTING FLOW.	9
FIGURE 2-1. FLASH CLOCK PARAMETERS.	12

LIST OF TABLES

TABLE 1-1. SELECT OPERATION MODE.....	7
TABLE 2-1. PULL-UP/DOWN SETTINGS.	11
TABLE 2-2. REGISTER SETTINGS AFTER ON-CHIP ROM BOOT.	12
TABLE 2-3. PULL-UP/DOWN SETTINGS.	14
TABLE 3-1. RELATED DOCUMENT.	15

1 VERSION LIST

Description	Version	Date	Author
Introduction	V0.0.1	2024/08/05	Renny Hsu
Modify DIP switch setting.		2024/09/18	Renny Hsu

2 HASHTAG

#V4H

#QSPI

#SDR

#DDR

#Single Fast

#Quad

1. Overview

Before the IPL (Initial Program Loader) and other firmware images were loaded from QSPI flash, choose QSPI flash operation mode by SW28:

Table 1-1. Select operation mode.

SW28				
Pin5	Pin6	Pin7	Pin8	Operation Mode
ON	OFF	ON	ON	Serial Flash boot at single read 40MHz using DMA
ON	OFF	OFF	ON	Serial Flash boot at DDR read 80MHz using DMA

1.1. Serial Flash boot at single read 40MHz using DMA

In this option, QSPI flash works in external address space read mode. By external address space read mode, a read access to the SPI multi I/O bus space is converted into the data read protocol. Further, the commands, optional commands, option data, and dummy cycle issued for reading external device can be modified using registers.

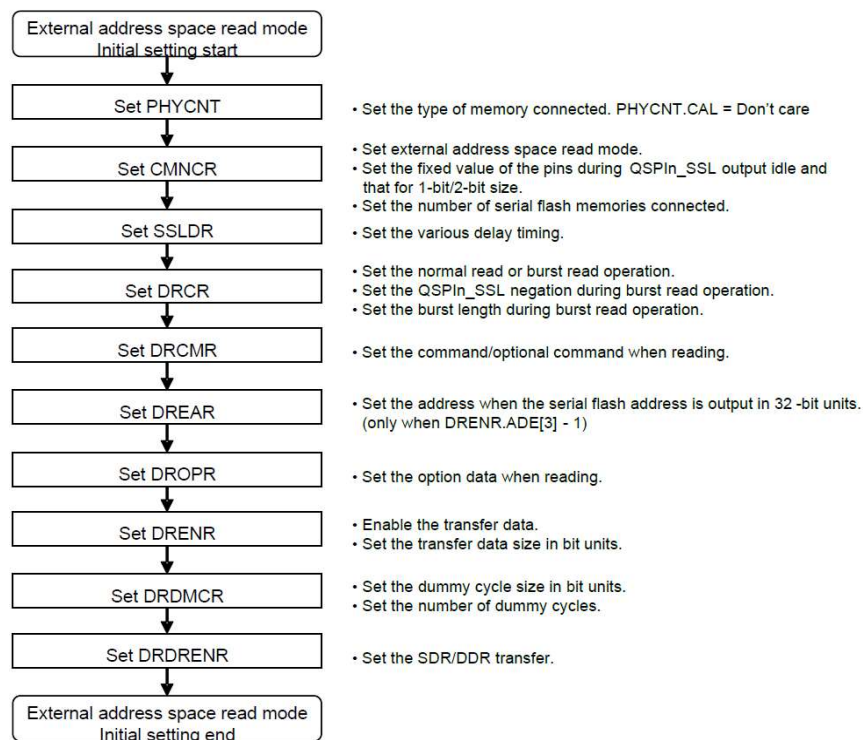


Figure 1-1. External address space read mode initial flow.

1.2. Serial Flash boot at DDR read 80MHz using DMA

In this option, QSPI flash works in manual operating mode and can carry out an arbitrary serial transfer operation by using the register settings.

The transfer format is determined based on the common control register (CMNCR), SSL delay register (SSLDR), manual mode control register (SMCR), manual mode command setting register (SMCMR), manual mode address setting register (SMADR), manual mode option setting register (SMOPR), and manual mode enable setting register (SMENR), manual mode read data register (SMRDR), manual mode write data register (SMWDR), manual mode dummy cycle setting register (SMDMCR), and manual mode DDR enable register (SMDRENDR).

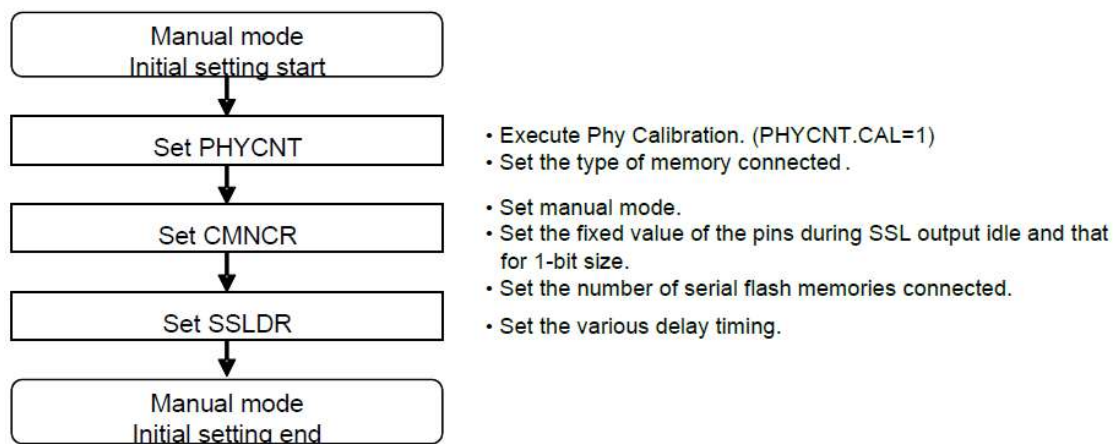


Figure 1-2. Manual operating mode initial flow.

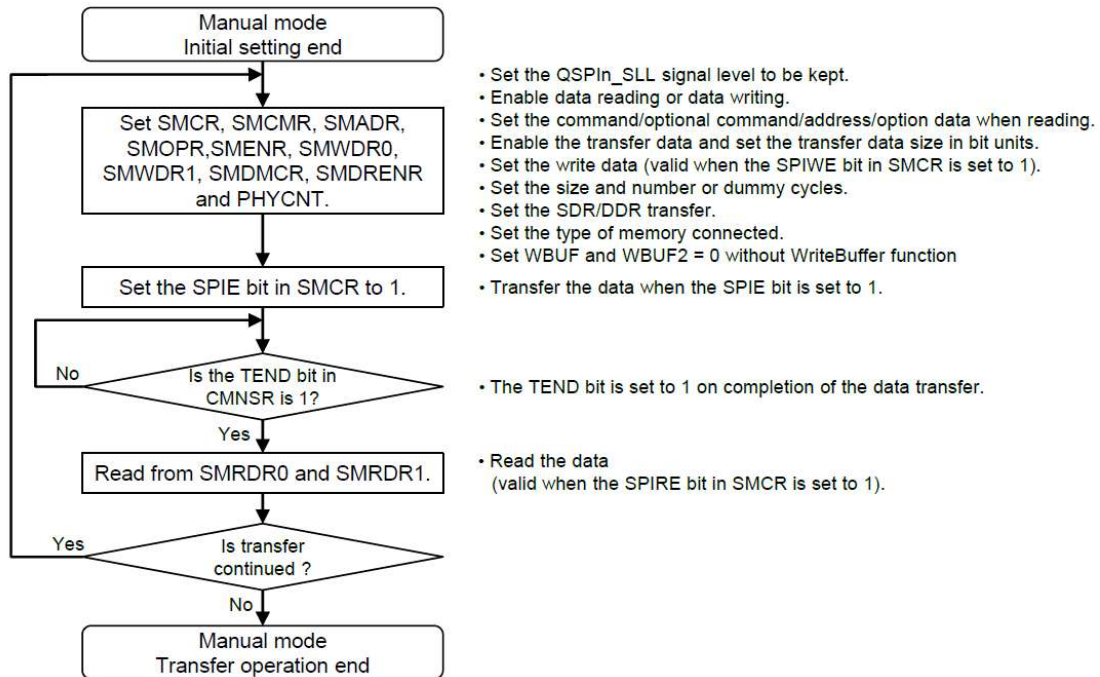


Figure 1-3. Manual operating mode data transfer setting flow.

2.Setup Step

2.1. Serial Flash boot at single read 40MHz using DMA

2.1.1. Setting boot parameter in sa0.c

```
/* SA0 */  
/* boot parameter */  
__attribute__((section(".sa0_boot"))) const uint32_t boot[] =  
{  
    [0] = BOOT_PARAM,  
    [1] = 0xFFFFFFFFU, /* 40MHz Single fast read */  
};
```

2.1.2. Following qspi_sdr_transfer_mode() in rpc.c

```
/* PHY calibration */  
set_strtim(STRTIM_SMALLEST);  
reg = mem_read32(RPC_PHYCNT);  
reg |= RPC_PHYCNT_CAL;  
mem_write32(RPC_PHYCNT, reg);  
  
/* External Address Space Read Mode */  
reg = mem_read32(RPC_CMNCR);  
reg &= ~(RPC_CMNCR_MD);  
mem_write32(RPC_CMNCR, reg);  
  
/* Read cache Flash */  
reg = mem_read32(RPC_DRCR);  
reg |= RPC_DRCR_RCF;  
mem_write32(RPC_DRCR, reg);  
  
/* 32bit address read command*/  
reg = mem_read32(RPC_DRCMR);  
reg &= ~(RPC_DRCMR_CMD_MASK | RPC_DRCMR_OCMD_MASK);  
reg |= (command << RPC_SMCMR_CMD_SHIFT);  
mem_write32(RPC_DRCMR, reg);  
  
/* Extended external address valid range is [25:0]*/
```

```
reg = mem_read32(RPC_DREAR);
reg &= ~(RPC_DREAR_EAV_MASK | RPC_DREAR_EAC_MASK);
reg |= RPC_DREAR_EAC_EXT_ADDR_25BIT;
mem_write32(RPC_DREAR, reg);

/* output command is 32bit width */
reg = mem_read32(RPC_DRENr);
reg &= ~(RPC_DRENr_CDE_ENABLE | RPC_DRENr_ADE_MASK);
reg |= (RPC_DRENr_CDE_ENABLE | RPC_DRENr_ADE_32BIT_ADDR);
mem_write32(RPC_DRENr, reg);
```

2.1.3. Pull-up/down settings.

Table 2-1. Pull-up/down settings.

SW28	Descriptions	QSPI0_IO2/ QSPI0_IO3	QSPI1_SPCLK	QSPI1_MOSI_IO0/ QSPI1_MISO_IO1/ QSPI1_IO2/QSPI1_IO3
ON OFF ON ON	Serial Flash 40MHz	Pull-up	Pull-down	Pull-up

2.2. Serial Flash boot at DDR read 80MHz using DMA

2.2.1. Setting boot parameter in sa0.c

```
/* SA0 */
/* boot parameter */
__attribute__((section(".sa0_boot"))) const uint32_t boot[] =
{
    [0] = BOOT_PARAM,
    [1] = 0xFFFFFFFF00U, /* 80MHz Single fast read */
};
```

Bit	Bit Name	Initial Value	R/W	Description
31 to 8	-	All 0	-	Reserved
7 to 0	CLKSELR	-	-	QSPI Flash clock setting at MD[4:1]=B'0110 B'00000101: 80MHz Quad I/O B'00000100: 133MHz Quad I/O B'00000011: 80MHz Quad output read B'00000010: 133MHz Quad output read B'00000001: Reserved B'00000000: 80MHz Single fast read Others: 40MHz Single fast read Octal SPI Flash clock setting at MD[4:1]=B'0111 B'00000101: 160MHz Octal SPI for Macronix's products B'00000100: 80MHz Octal SPI for Macronix's products B'00000011: 160MHz Octal SPI for Micron's products B'00000010: 80MHz Octal SPI for Micron's products Others: Setting prohibited

Figure 2-1. Flash Clock parameters.

Table 2-2. Register settings after on-chip ROM boot.

Boot Parameter	CLKSELR	B'0000 0000
Specification	Operating clock	80MHz
	Bit width(Command-Address-Data)	1-1-1
	Command	H'0B
	Address width	24 bit
	Dummy cycle	8 cycle
Register Setting	CMNCR	H'0155 5300
	DRCR	H'001F 0101
	DRCMR	H'00ED 0000
	DRENCR	H'0202 C700
	DRDMCR	H'0000 0008
	DRDRENCR	H'0000 0101
	PHYCNT	H'0807 8260
	PHYOFFSET1	H'3151 1144
	PHYINT	H'0606 0002
	RPCCKCR	H'0000 0013

2.2.2. Following rpc_init () in rpc.c

```

read_any_register_qspi_flash(S25FS512S_CR1V_ADDR, &statusReg);
//ERROR("CR1V(0x%08X) = 0x%02X\n", S25FS512S_CR1V_ADDR, statusReg);

/* write ENABLE */
write_command_qspi_flash((S25FS512S_WRITE_ENABLE) <<
RPC_SMCMD_CMD_SHIFT);

```

```
/* set quad I/O mode */
statusReg |= (0x01 << 1);
write_any_register_qspi_flash(S25FS512S_CR1V_ADDR, statusReg);
```

2.2.3. Following qspi_dds_transfer_mode_quad_io_read () in rpc.c

```
/* PHY calibration */
set_strtim(STRTIM_SMALLEST);
reg = mem_read32(RPC_PHYCNT);
reg |= RPC_PHYCNT_CAL;
mem_write32(RPC_PHYCNT, reg);

/* External Address Space Read Mode */
reg = mem_read32(RPC_CMNCR);
reg &= ~(RPC_CMNCR_MD | RPC_CMNCR_MOIIO_MASK |
RPC_CMNCR_BSZ_MASK);
reg |= (RPC_CMNCR_MOIIO0_HI_Z | RPC_CMNCR_MOIIO1_HI_Z |
RPC_CMNCR_MOIIO2_HI_Z | RPC_CMNCR_MOIIO3_HI_Z);
mem_write32(RPC_CMNCR, reg);

/* Read cache Flash */
reg = mem_read32(RPC_DRCR);
reg &= ~(RPC_DRCR_RBE_RBURST_MASK);
reg |= (RPC_DRCR_SSLN_NEGATE | RPC_DRCR_RBE_RBURST_1DATA |
RPC_DRCR_RCF | RPC_DRCR_RBE_ENABLE);
mem_write32(RPC_DRCR, (reg | RPC_DRCR_RCF));

/* 32bit address read command*/
reg = mem_read32(RPC_DRCMR);
reg &= ~(RPC_DRCMR_CMD_MASK | RPC_DRCMR_OCMD_MASK);
reg |= (command << RPC_SMCMD_CMD_SHIFT);
mem_write32(RPC_DRCMR, reg);

/* Extended external address valid range is [25:0]*/
reg = mem_read32(RPC_DREAR);
reg &= ~(RPC_DREAR_EAV_MASK | RPC_DREAR_EAC_MASK);
reg |= RPC_DREAR_EAC_EXT_ADDR_25BIT;
mem_write32(RPC_DREAR, reg);

/* output command is 32bit width */
reg = mem_read32(RPC_DRENDR);
reg &= ~(RPC_DRENDR_CDB_MASK | RPC_DRENDR_ADB_MASK |
RPC_DRENDR_DRDB_MASK
| RPC_DRENDR_DME_MASK | RPC_DRENDR_CDE_ENABLE |
```

```

RPC_DRENDR_ADE_MASK);
reg |= (RPC_DRENDR_CDB_1BIT      | RPC_DRENDR_ADB_4BIT      |
RPC_DRENDR_DRDB_4BIT
      |   RPC_DRENDR_DME_ENABLE | RPC_DRENDR_CDE_ENABLE |
RPC_DRENDR_ADE_24BIT_ADDR);
mem_write32(RPC_DRENDR, reg);

/* Set Dummy Cycle */
mem_write32(RPC_DRDMCR, (uint32_t)RCAR_QSPI_DDR_DUMMY_CYCLE - 1U);

/* Specifies DDR transfer of the data read. */
reg = mem_read32(RPC_DRDRENDR);
reg |= (RPC_DRDRENDR_DRDRE | RPC_DRDRENDR_ADDRE);
mem_write32(RPC_DRDRENDR, reg);

/* timing adjustment in DDR read operation. */
reg = mem_read32(RPC_PHYOFFSET1);
reg &= ~(RPC_PHYOFFSET1_DDRTMG_MSK);
reg |= RPC_PHYOFFSET1_DDRTMG_DDR;
mem_write32(RPC_PHYOFFSET1, reg);

/* Change PHYCNT.DDRCAL */
reg = mem_read32(RPC_PHYCNT);
reg |= RPC_PHYCNT_PHYMEM_SPI_DDR;
//reg &= ~RPC_PHYCNT_CAL;
reg |= RPC_PHYCNT_HS;
reg |=   RPC_PHYCNT_DDRCAL;
mem_write32(RPC_PHYCNT, reg);

//adjust_strobe_timing();

/* Set data burst length to 256 byte */
reg = mem_read32(RPC_DRCCR);
reg &= 0xFFFFFFF;
reg |= RPC_DRCCR_RBE_RBURST_32DATA;
mem_write32(RPC_DRCCR, reg);

```

2.2.4. Pull-up/down settings

Table 2-3. Pull-up/down settings.

SW28	Descriptions	QSPI0_IO2/ QSPI0_IO3	QSPI1_SPCLK	QSPI1_MOSI_IO0/ QSPI1_MISO_IO1/ QSPI1_IO2/QSPI1_IO3
ON OFF OFF ON	Serial Flash 40MHz To 80MHz (CLKSELR=0)	Pull-up	Pull-down	Pull-up

3. References

The following table shows the document related to this function.

Table 3-1. Related Document.

Number	Issue	Title	Version
[1]	Renesas Electronics	R-Car V4H Series User's Manual: Hardware	Rev.0.82
[2]	Infineon	Infineon_S25HSxxxT_S25HLxxxT.pdf	Rev.AD
[3]	Infineon	Infineon_S25FS512S_512_Mb_1_8_V_Serial_Peripheral_-3363590.pdf	Rev.N
[4]	Renesas Electronics	R-CarV4H_UM_112_RPC_r0p81.xlsx	Rev0.81